

ASHMIT ANAND

MERN Stack | Full-Stack Developer

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PROFESSIONAL SUMMARY

I'm interested in building practical technology that solves real problems — from AI-powered systems and automation workflows to scalable full-stack applications. I enjoy turning ideas into usable products and exploring the space where engineering, creativity, and product thinking meet.

EDUCATION

Bachelor of Technology — Computer Science & Engineering

Haldia Institute of Technology

2024 – 2028

Haldia, West Bengal

Higher Secondary School (Class XII)

Bridgeford School

2022 – 2024

Ranchi, Jharkhand

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript

Web Development: HTML, CSS, React.js, Node.js, Express.js, REST API Design

Databases: MongoDB, MySQL, Supabase

AI / ML: NumPy, Pandas, Scikit-learn, ML, DL, CNN, Model Evaluation, Rag-Systems

Generative AI & NLP: NLP and ML pipelines, Agentic AI, OpenAI APIs

Tools & Platforms: GitHub, Claude, VS Code, Google Colab

PROJECTS

Deep-AE — AI-Powered Acoustic Structural Crack Detection System

AI/ML | Python, PyTorch, Streamlit, Librosa, NumPy, Matplotlib

- **Engineered** an end-to-end ML pipeline for acoustic structural health monitoring, classifying crack-related emissions via CNN-based audio classification and Mel Spectrogram analysis.
- **Designed** a CNN architecture for acoustic crack classification, integrating audio signal preprocessing, spectrogram generation, and real-time WAV/MP3/FLAC prediction with confidence analytics.
- **Developed** an interactive AI monitoring dashboard using Streamlit with live prediction workflows and Plotly-powered confidence visualization, enabling intuitive structural health assessment.
- **Implemented** a modular audio preprocessing pipeline with Librosa, achieving robust feature extraction and seamless deep learning integration across diverse audio input formats.

AI YouTube Thumbnail Style Cloner — Multi-Agent Thumbnail Generation System

Full-Stack | Python, Streamlit, OpenAI API, Google Gemini API, YouTube Data API

- **Orchestrated** an end-to-end AI pipeline to scrape YouTube thumbnails and extract visual style patterns for automated thumbnail generation.
- **Built** a modular multi-agent workflow integrating style analysis, prompt generation, and image synthesis.
- **Created** an interactive Streamlit UI for real-time thumbnail generation with A/B variations and download support.
- **Integrated** error handling and fallback mechanisms to ensure reliable execution across API failures.

StayHouse — Full-Stack Accommodation Booking Platform

Full-Stack | Node.js, Express.js, MongoDB, React.js

- **Designed** an Airbnb-inspired full-stack platform with end-to-end CRUD workflows for property listings, user accounts, and booking management.
- **Programmed** and implemented RESTful APIs with Express.js and MongoDB to handle concurrent booking logic, user authentication, and property filtering.
- **Enhanced** database query performance, achieving sub-200ms average response times for listing search and retrieval operations.
- **Applied** a responsive front-end with React.js, improving user engagement through component-driven UI architecture and dynamic state management.

ACHIEVEMENTS & ACTIVITIES

- Participated in hackathons and built real-world projects, including full-stack web apps and AI-based tools.
- Designed and built 3 end-to-end production-ready projects.
- Actively solve Data Structures and Algorithms (DSA) problems, focusing on logic building and optimization.
- Maintain active GitHub projects and contribute to open-source repositories, focusing on practical improvements and features.